NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

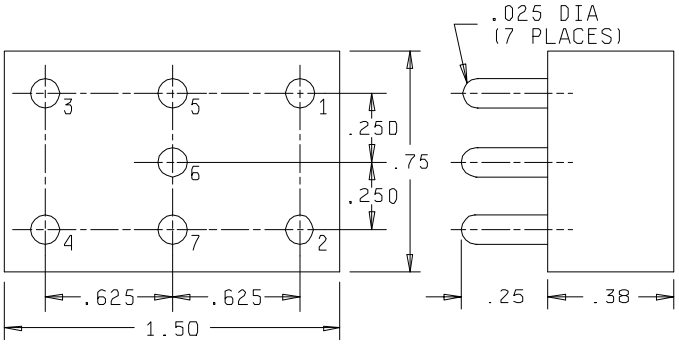
MIL-PRF-27/27
15 September 1975

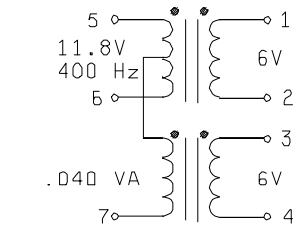
PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN, TF5V03ZZ314

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformers described herein shall consist of this document and the latest issue of Specification MIL-T-27.





CIRCUIT DIAGRAM AND MARKING

INCHES	MM
.025	0.64
.250	6.35
.38	9.65
.625	15.88
.75	19.05
1.50	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Tolerance on terminal position dimensions is ±.005 (0.13 mm) and ±.02 (0.51 mm) for remaining dimensions.
4. Marking shall be on the top of the case.
5. Terminals (5-6-7) may be to synchro input and terminals (1-2-3) may be to resolver output.

FIGURE 1. Dimensions and configuration.

FSC 5950

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REQUIREMENTS (When numbers in parentheses, i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings
Primary voltage
(5-6-7) - 11.8 volts, 400 Hz, 0.0035 ampere maximum
Secondary voltage
(1-2) - 6 volts, 6 mA rms
(3-4) - 6 volts, 6 mA, rms
Working voltage
(5-6-7) - 18 volts maximum
(1-2) (3-4) - 18 volts maximum
Volt ampere .040
Design and construction
Dimensions and configuration See figure 1
Duty cycle Continuous.
Case Encapsulated
Material Diallyl phthalate, glass filled
Terminals Pin type, copper clad, steel, tinned, copper weld
Terminal height .25 inch
Weight .20 grams.
Operating temperature range: -55°C to 155°C

Terminal strength MIL-STD-202, method 211, test condition A, 2 pounds

Dielectric withstanding voltage (at sea level) 50 volts rms

Electrical characteristics
No load - With 11.8 volts, 400 Hz across (5-6-7), current in (5-6-7) shall not exceed 3 mA and the power in (5-6-7) shall not exceed 5 milliwatts
Voltage across (1-2) and (3-4) shall be 6 volts, respectively
Rated load - with 11.8 volts, 400 Hz, .0035 ampere, maximum across (5-6-7)
Voltage across (1-2) and (3-4) shall be 6 volts, respectively
Shaft angle transformation accuracy 5 minutes of arc synchro input to resolver output
Temperature rise 5°C maximum

Marking location See figure 1
Part number M27/27-01

Custodians
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - EL, MI, SG
Navy - EC, SH, OS
Air Force - 11, 17, 80
DSA - ES

User activities
Army - MU, WC, SG, ML
Navy - MC, AS, SH, OS
Air Force - 19

Preparing activity
Army - EL

Agent
DSA - ES

(Project 5950-0496)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: MIL-T-27727 TRANSFORMER, POWER, STEP-DOWN, TRANSISTOR

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   YES NO (If "yes" in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional) | DATE

DD FORM 1 JAN 56 1426 REPLACES PREVIOUS EDITION ESC-FM 1068-68

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